MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2026

FOURTH SEMESTER

MINOR - MG4DSCBTG202 TISSUE CULTURE TECHNIQUES

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A – One word answer - Answer any 10 questions. Each question carries 2 marks

1. Name any two chemical sterilants.	[K] /CO2
2. Who is the father of plant tissue culture?	[K] /CO1
3. What is the role of activated charcoal in PTC media?	[U] / CO2
4. Define macronutrients	[U] / CO2
5. What type of culture is used to produce virus free plants?	[K] / CO3
6. Define callus.	[U] / CO3
7 Give two examples of secondary metabolites produced by plants.	[K] / CO3
8. Name any two single cell culturing techniques in Plant Tissue Culture.	[U] / CO3
9. What is continuous cell line?	[U] / CO3
10. Name any two animal cell culture media.	[U] / CO1
11. Define transformed cell.	[U] / CO4
12. What is monoclonal antibody?	[U] / CO4
	(10X2=20)

Part B- Short essay- Answer any 5 questions. Each question carries 4 marks

13. Describe the history of plant biotechnology.	[A] / CO1
14. What are the applications of suspension culture?	[A] / CO3
15. Write notes on the composition and preparation of MS media	[U] / CO2
16. Explain somaclonal variation.	[U] / CO3

17. What are the different modes of regeneration of plantlets in plant tissue culture. [R] / CO3

18. What are stem cells? Describe the applications of stem cell culture. [R] / CO4

19.Write in detail the significance of animal cell culture in medical and cancer research[A] / CO4

(5X4=20)

Part C- Essay- Answer any 1 question. Each question carries 10 marks

20. Write a detailed account on the large scale culture of animal cell lines [A] / CO4.

21. Write an essay on the applications of plant cell culture. [A] /CO3

(1X10=10)

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